Naïve Baye's Classifier

Bayes Theorm

- **Bayes' theorem** describes the probability of an event, based on prior knowledge of conditions that might be related to the event.
- For example, if cancer is related to age, then, using

Bayes’ theorem, a person’s age can be used to more accurately assess the probability that they have cancer, compared to the assessment of the probability of cancer made without knowledge of the person's age.

Understanding Bayesian methods

- Classifiers based on Bayesian methods utilize training data to calculate an **observed probability** of each outcome based on the **evidence** provided by feature values.
- When the classifier is later applied to unlabeled

data, it uses the observed probabilities to predict the most likely class for the new features.

- It's a simple idea, but it results in a method that often has results on par with more sophisticated algorithms.

Bayes’ theorem (named after Rev. Thomas Bayes 1702-1761), is based on this

- *It states, for two events A & B, if we know the conditional probability of B given A and the probability of B, then it’s possible to calculate the probability of A given B.*

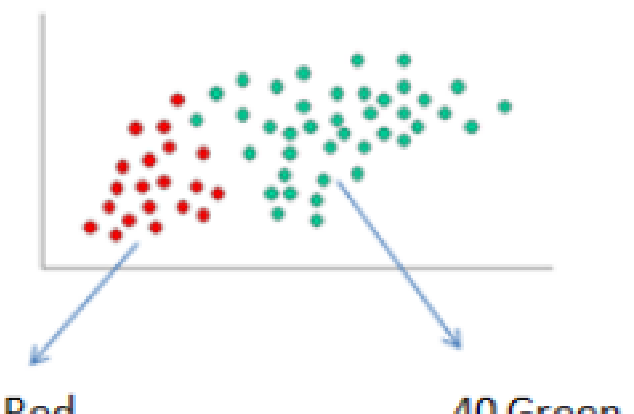
$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

where A and B are events and $P(B) \neq 0$.

- $P(A \mid B)$ is a conditional probability: the likelihood of event A occurring given that B is true.
- $P(B \mid A)$ is also a conditional probability: the likelihood of event B occurring given that A is true.
- $P(A)$ and $P(B)$ are the probabilities of observing A and B independently of each other; this is known as the marginal probability.

- Below is one simple way to explain the Bayes rule. The task is to identify the color of a newly-observed dot.

Since there are twice as many GREEN objects
as RED, it is reasonable to believe that a new
case (which hasn't been observed yet) is



twice as likely to have membership with GREEN rather than RED. In the Bayesian analysis, this belief is known as **the prior probability**.

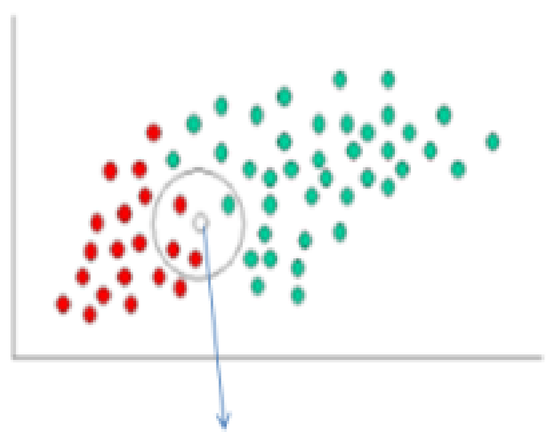
Prior probabilities are based on previous experience, in this case the percentage of GREEN and RED objects, and often used to predict outcomes before they actually happen.

Since there is a total of 60 objects, 40 of which are GREEN and 20 RED, our prior probabilities for class membership can be written as as:

$$Prior\ Probability\ of\ GREEN = \frac{number\ of\ GREEN\ objects}{total\ number\ of\ objects}$$

$$Prior\ Probability\ of\ RED = \frac{number\ of\ RED\ objects}{total\ number\ of\ objects} =$$

Having formulated our prior probability, we are now ready to classify a new object. Since the objects are well clustered, it is reasonable to assume that the more GREEN (or RED) objects in the vicinity of X, the more likely that the new cases belong to that particular color.



LIKELIHOOD OF THIS BEING RED = (NO. OF RED IN THE VICINITY / TOTAL NO OF RED)

- To measure this likelihood, we draw a circle around X which encompasses a number (to be chosen a priori) of points irrespective of their class labels.
- Then we calculate the number of points in the circle belonging to each class label. From this **we calculate the likelihood**:
- From the illustration above, it is clear that the likelihood of X given GREEN is smaller than Likelihood of X given RED, since the circle encompasses 1 GREEN object and 3

RED ones.

- Although the **prior probabilities indicate that X may belong to GREEN** (given that there are twice as many GREEN compared to RED) the **likelihood indicates otherwise; that the class membership of X is RED** (given that there are more RED objects in the vicinity of X than GREEN).
- **In the Bayesian analysis, the final classification is produced by combining both sources of information, i.e., the prior and the**

likelihood, to form a posterior probability using Bayes' rule.

Posterior Probability of GREEN= Prior Probability of GREEN×Likelihood of GREEN= (40/60)×(1/40)

Posterior Probability of RED=Prior Probability of RED×Likel ihood of RED=(20/60)×(3/20)

What is Naive Bayes algorithm?

- It is a classification technique based on Bayes' Theorem with an assumption of independence among predictors.

- In simple terms, a Naive Bayes classifier assumes that the presence of a particular feature in a class is unrelated to the presence of any other feature.
- For example, a fruit may be considered to be an apple if it is red, round, and about 3 inches in diameter.
- Even if these features depend on each other or upon the existence of the other features, all of these properties independently contribute to the probability that this fruit is an apple and that is why it is known as '**Naive'.**

- Naive Bayes model is easy to build and particularly useful for very large data sets. Along with simplicity, Naive Bayes is known to outperform even highly sophisticated classification methods.
- Bayes theorem provides a way of calculating posterior probability $P(c|x)$ from $P(c)$, $P(x)$ and $P(x|c)$.

Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)\,P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

- *P*(*c*|*x*) is the posterior probability of *class* (c, *target*) given *predictor* (x, *attributes*).
- *P*(*c*) is the prior probability of *class*.
- *P*(*x*|*c*) is the likelihood which is the probability of *predictor* given *class*.

- $P(x)$ is the prior probability of *predictor*.

Example: Picnic Day

- You are planning a picnic today, but the morning is cloudy
- Oh no! 50% of all rainy days start off cloudy!
- But cloudy mornings are common (about 40% of days start cloudy)
- And this is usually a dry month (only 3 of 30 days tend to be rainy, or 10%)

What is the chance of rain during the day?

- We will use Rain to mean rain during the day, and Cloud to mean cloudy morning.
- **The chance of Rain given Cloud is written P(Rain|Cloud)**
- So let's put that in the formula:

$$P(\text{Rain}|\text{Cloud}) = \frac{P(\text{Rain})\ P(\text{Cloud}|\text{Rain})}{P(\text{Cloud})}$$

P(Rain) is Probability of Rain = 10%
P(Cloud|Rain) is Probability of Cloud, given that Rain happens = 50%
P(Cloud) is Probability of Cloud = 40%

$$P(\text{Rain}|\text{Cloud}) = \frac{0.1 \times 0.5}{0.4} = .125$$

Or a 12.5% chance of rain. Not too bad, let's have a picnic!

How Naive Bayes algorithm works?

• We have a training data set of weather and corresponding target variable ‘Play’ (suggesting possibilities of playing). • Now, we need to classify whether players will play or not based on weather condition.

Let’s follow the below steps to perform it.

• **Step 1**: Convert the data set into a frequency table • **Step 2**: Create Likelihood table by finding the probabilities like Overcast probability = 0.29 and probability of playing is 0.64.

How Naive Bayes algorithm works?

Whether	Play
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes
Rainy	No
Overcast	Yes
Sunny	No
Sunny	Yes
Rainy	Yes
Sunny	Yes
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table

Whether	No	Yes
Overcast		4
Sunny	2	3
Rainy	3	2
Total	5	9

Likelihood Table 1

Whether	No	Yes		
Overcast		4	=4/14	0.29
Sunny	2	3	=5/14	0.36
Rainy	3	2	=5/14	0.36
Total	5	9		
	=5/14	=9/14		
	0.36	0.64		

Likelihood Table 2

Whether	No	Yes	Posterior Probability for No	Posterior Probability for Yes
Overcast		4	0/5=0	4/9=0.44
Sunny	2	3	2/5=0.4	3/9=0.33
Rainy	3	2	3/5=0.6	2/9=0.22
Total	5	9		

Step 3: Now, use Naive Bayesian equation to calculate the posterior probability for each class. The class with the highest posterior probability is the outcome of prediction.

How Naive Bayes algorithm works?

- **Problem:** Players will play if weather is sunny. Is this statement is correct?
- We can solve it using method of posterior probability. • **P(Yes | Sunny) = P(Sunny | Yes) * P(Yes) / P (Sunny)** • Here we have P (Sunny |Yes) = 3/9 = 0.33, P(Sunny) = 5/14 = 0.36, P(Yes)= 9/14 = 0.64
- Now, P (Yes | Sunny) = 0.33 * 0.64 / 0.36 = 0.60, which has higher probability.
- Naive Bayes uses a similar method to predict the probability of different class based on various attributes. • This algorithm is mostly used in text classification and with

problems having multiple classes.

How to build a basic model using Naive Bayes in Python

Here are three types of Naive Bayes model under the scikit-learn library:

- **Gaussian:** Gaussian Naïve Bayes is used when we assume all the continuous variables associated with each feature to be distributed according to **Gaussian Distribution.** Gaussian Distribution is also called Normal distribution.

- **Multinomial:** It is used for discrete counts. For example, let's say, we have a text classification problem. Here we can consider Bernoulli trials which is one step further and instead of "word occurring in the document", we have "count how often word occurs in the document", you can think of it as "number of times outcome number x_i is observed over the n trials”.

•**Bernoulli**: The binomial model is useful if your feature vectors are binary (i.e. zeros and ones). One application would be text classification with ‘bag of words’ model where the 1s & 0s are “word occurs in the document” and “word does not occur in the document” respectively.

What are the Pros and Cons of Naive Bayes? *Pros:*

- It is easy and fast to predict class of test data set. It also perform well in multi class prediction
- When assumption of independence holds, a Naive Bayes classifier performs better compare to other models like logistic regression and you need less training data.
- It perform well in case of categorical input variables compared to numerical variable(s). For numerical variable, normal distribution is assumed (bell

curve, which is a strong assumption).

Cons:

- If categorical variable has a category (in test data set), which was not observed in training data set, then model will assign a 0 (zero) probability and will be unable to make a prediction. This is often known as "**Zero Frequency**". To solve this, we can use the smoothing technique. One of the simplest smoothing techniques is called **Laplace estimation**.

• On the other side naive Bayes is also known as a bad estimator, • Another limitation of Naive Bayes is the assumption of independent predictors. In real life, it is almost impossible that we get a set of predictors which are completely independent.